

Highlights

Exploring Decoder Computation and Redundancy in 3D Medical Image Segmentation: A Voxel-Level Characterization Study

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- Characterizes spatial contribution of 3D decoder computation in medical segmentation.
- Formalizes voxel correctness activity versus directional net gain divergence.
- Reveals boundary-localized bidirectional activity and high gain concentration ($K_{80} \leq 5\%$).
- Uncovers Activity–Direction Gap: uncertainty signals activity, not net utility.
- Proposes a stratified clinical deployment paradigm for efficient 3D decoders.

Exploring Decoder Computation and Redundancy in 3D Medical Image Segmentation: A Voxel-Level Characterization Study

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Abstract

Modern 3D medical image segmentation architectures increasingly incorporate heavy multi-resolution decoders, implicitly assuming that dense decoding computation is universally necessary across the volumetric grid. Conversely, efficient alternatives simplify these pathways based solely on global accuracy metrics. However, where, how much, and under what spatial conditions additional decoder computation genuinely improves segmentation correctness remains uncharacterized. We present a systematic characterization framework for 3D decoder computation, evaluating matched full-efficient decoder pairs across diverse architectural backbones, multi-modality datasets, controlled 2×2 factorial factor attributions, and two distinct decoder-efficiency mechanisms (capacity removal and a capacity-preserving factorization). Our framework formalizes the separation between two quantities that aggregate metrics conflate: *correctness activity* (where the decoder alters a voxel’s correctness) and *directional net gain* (the signed balance of corrections and degradations). We demonstrate that the two diverge sharply: across architectures and modalities, added decoder capacity exhibits high gross activity yet is nearly net-neutral, with corrections and degradations largely cancelling out over the volume. The residual gain is highly concentrated ($K_{80} \leq 5\%$) and confined to a narrow 0–2 mm anatomical boundary margin. Crucially, while predictive uncertainty reliably locates *where* correctness changes occur (Location AUROC ≥ 0.86), it provides weak directional discrimination for *whether* predictions improve (Direction AUROC ≈ 0.52 – 0.59), establishing that prediction difficulty and computational utility are related but distinct. Clinically, these findings provide a stratified deployment paradigm: globally lightweight decoders can be safely trusted for real-time intraoperative guidance and bulk organ segmentation, whereas heavy decoding capacity can be sparsely allocated to critical anatomical margins to safeguard high-risk surgical and radiotherapeutic decisions.

Keywords: 3D Medical Image Segmentation, Efficient Deep Learning, Uncertainty Quantification, Decoder Computation, Model Characterization

1. Introduction

Precise 3D medical image segmentation is essential for clinical applications such as image-guided surgical planning, volumetric tumor monitoring, and targeted radiation therapy. In deep learning, encoder-decoder networks based on the U-Net paradigm [1, 2] remain the foundational standard. Recent research has primarily focused on scaling encoder expressiveness through vision transformers, large convolutional kernels, and self-supervised pre-training [3, 4, 5, 6]. In parallel, modern 3D architectures retain heavy multi-resolution decoders equipped with dense upsampling stages, complex skip-fusion connections, and spatial refinement blocks. Because 3D volumetric scans scale cubically in resolution ($O(N^3)$), these multi-scale decoders routinely account for over 40% of total inference FLOPs, posing severe memory and latency bottlenecks for real-time intraoperative deployment. This prevailing architectural trajectory implicitly assumes that dense, high-capacity decoding is universally necessary across every voxel in the 3D volume.

Conversely, emerging lightweight models attempt to mitigate this computational burden by simplifying decoding pathways, such as EffiDec3D [7] and EfficientMedNeXt [8]. Yet both the heavy and the lightweight camps are judged on the same footing: standard evaluation paradigms assess decoder designs almost exclusively through aggregate metrics such as the global Dice similarity coefficient or Hausdorff distance. Herein lies an information gap. A single volume-averaged scalar collapses millions of independent voxel-level decisions into one number, discarding precisely the spatial micro-dynamics it is meant to explain—which voxels the extra computation flips, in which direction, and under what anatomical conditions. A decoder that rewrites thousands of voxels near anatomical boundaries and one that leaves the volume nearly untouched can report an identical Dice score. This gap is not merely academic: the boundary voxels that aggregate metrics dilute are exactly the ones that determine clinical outcomes—tumor margins that dictate resection extent, vessel walls that constrain catheter navigation, and organ interfaces that bound radiation dose. A characterization that resolves *where* and *whether* decoder computation helps therefore speaks directly to whether a lighter, faster model can be trusted at the anatomical sites where errors carry the greatest

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clinical cost. Consequently, a fundamental scientific question remains unexamined: **How does extra decoder computation deliver spatially-varying performance gains for volumetric medical segmentation?**

To address this question, we systematically characterize decoder computation in 3D medical image segmentation models. Rather than proposing a new network architecture, we treat computation characterization as a distinct empirical approach. The main contributions of this study are:

1. We introduce a voxel-level framework that separates decoder correctness activity from directional net gain.
2. We characterize decoder contribution in terms of spatial heterogeneity, positive-gain concentration, and inference-time predictability.
3. We provide controlled evidence across architectures, datasets, and two distinct decoder-efficiency mechanisms—capacity removal and a capacity-preserving factorization—showing the characterization does not depend on how the decoder is made efficient, and we establish retrospective opportunity bounds for future utility-aware adaptive decoding.

The remainder of this paper is organized as follows: Section 2 reviews related literature and highlights key gaps. Section 3 details our characterization framework and metrics. Section 4 describes our experimental setup and evaluation protocols. Section 5 presents our observational findings and synthesizes their architectural, statistical, and clinical implications, and Section 6 concludes the paper.

2. Related Work

2.1. Decoder Design in 3D Medical Image Segmentation

The encoder-decoder architecture introduced by U-Net [1, 2] and V-Net [9] remains the central design pattern for 3D medical image segmentation. Frameworks such as nnU-Net [10] demonstrated the effectiveness of standardized training protocols combined with deep decoding pathways. Subsequent developments incorporated Transformer backbones and large-kernel convolutions to capture long-range contextual dependencies (e.g., TransUNet [11], UNETR [3], Swin UNETR [4], 3D UX-Net [5], and MedNeXt [6]). These modern architectures emphasize expressive feature extraction while maintaining dense multi-scale decoders. However, these designs are evaluated primarily through final segmentation accuracy and overall parameter count; how decoder capacity operates spatially across volume regions remains unexamined.

2.2. Efficient Segmentation Architecture Design

To mitigate the high computational cost of 3D volumetric convolutions, recent works have explored lightweight decoder designs. EffiDec3D [7] introduced systematic decoder pruning by reducing channel dimensions and eliminating high-resolution decoding stages, establishing trade-offs between parameter scale and Dice performance. Similarly, EfficientMedNeXt [8] simplified decoding stages before introducing residual

blocks. Other lightweight models achieve efficiency through hybrid transformer-convolution encoders, attention pruning, or custom lightweight modules (e.g., SegFormer [12], UNETR++ [13], Slim UNETR [14]). Nevertheless, efficient architecture studies infer redundancy indirectly from global performance curves, leaving unexamined whether the eliminated computation was spatially uniform or concentrated in specific anatomical sub-regions.

2.3. Model Characterization and Prediction Reliability

Existing characterization studies in medical image segmentation predominantly assess prediction reliability through uncertainty estimation, calibration, and failure detection [15, 16, 17, 18, 19]. In particular, recent work evaluates how uncertainty metrics reflect prediction errors during clinical deployment [20], or leverages Bayesian modeling for segmentation quality prediction [21]. While these approaches provide valuable quality assurance by estimating whether a model’s *static* output is reliable, they evaluate inference strictly as a fixed, single-pass process.

Our work bridges a distinct gap by shifting the perspective from static output reliability to the *marginal utility and spatial redundancy of decoder computation*. Rather than evaluating whether a static prediction is failure-prone, we systematically characterize where, how much, and under what spatial conditions allocating extra decoder capacity genuinely alters prediction correctness. This counterfactual computational lens establishes the empirical foundation for future utility-aware adaptive decoding.

3. Methods: A Characterization Framework

To understand the spatial and functional contribution of decoder computation, we establish a controlled observation framework (Figure 1). This section outlines the controlled decoder interventions, voxel-level correctness framework, rate definitions, spatial block partitioning, and characterization metrics.

3.1. Controlled Decoder Interventions

We analyze decoder capacity by isolating two generic architectural factors that govern volumetric decoding computation:

1. **Channel Capacity (C_{dec}):** The feature channel dimension across upsampling layers, controlling feature representation width ($C_{\text{full}} \rightarrow C_{\text{reduced}}$).
2. **High-Resolution Decoder Stages (RF):** The resolution scaling factor, controlling whether high-resolution upsampling stages (e.g., full resolution $1\times$) are retained (RF_1) or omitted (RF_2).

Crossing these two experimental factors forms a 2×2 factorial intervention design (Full E_0 , Channel-reduced, Stage-reduced, and Combined-efficient E_1). The specific channel-reduction ratios and stage-removal configurations follow the architecture-preserving decoder simplification strategy introduced by EffiDec3D [7]. Interventions occur strictly within

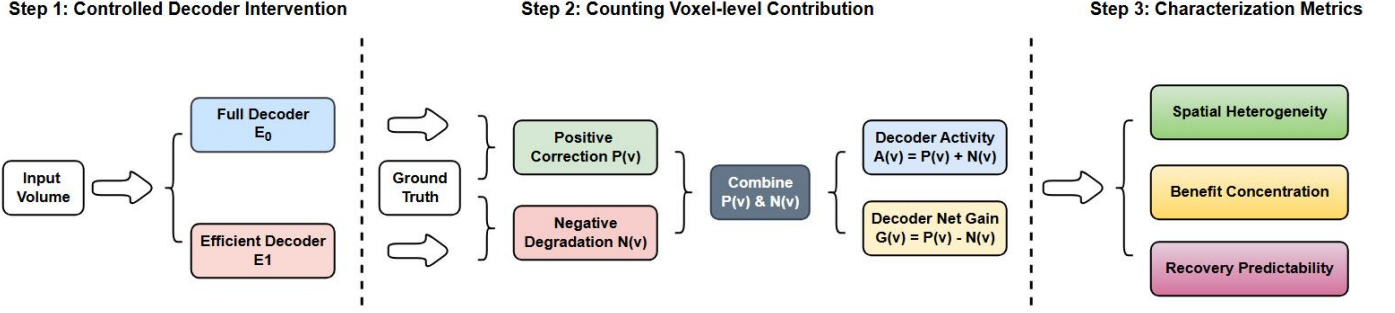


Figure 1: Overview of the decoder computation characterization framework, encompassing controlled decoder interventions (Step 1), voxel-level activity and net gain formalization (Step 2), and the three characterization axes (Step 3).

the decoder pathway (upsampling, skip fusion, and refinement) while keeping the feature encoder architecture untouched.

Choice of Intervention Template: We adopt the reduction principles of EffiDec3D as a controlled intervention template rather than treating EffiDec3D itself as the primary object of evaluation. Its decoder-localized and architecture-preserving modifications provide matched configurations in which channel capacity and high-resolution stage retention can be independently controlled. This enables spatial characterization of decoder contribution while minimizing confounding changes to the encoder family, prediction task, or training objective.

3.2. Voxel-Level Correctness Activity Framework

Let v denote a voxel within a 3D medical image volume V , with ground truth label $y(v)$, full decoder prediction $\hat{y}_f(v)$, and efficient decoder prediction $\hat{y}_e(v)$. Comparing predictions between matched E_0 and E_1 decoders yields two correctness-changing outcomes per voxel (Figure 2; see Supplementary Figures S1 and S2 for 3D UX-Net and MedNeXt-M-K3 visualizations):

- **Positive Correction ($P(v)$):** $P(v) = \mathbf{1}[\hat{y}_e(v) \neq y(v) \wedge \hat{y}_f(v) = y(v)]$, where the full decoder corrects an efficient baseline error.
- **Negative Degradation ($N(v)$):** $N(v) = \mathbf{1}[\hat{y}_e(v) = y(v) \wedge \hat{y}_f(v) \neq y(v)]$, where the full decoder alters a previously correct efficient prediction.

Defining voxel correctness indicators $C_f(v) = \mathbf{1}[\hat{y}_f(v) = y(v)]$ and $C_e(v) = \mathbf{1}[\hat{y}_e(v) = y(v)]$, we formalize two fundamental voxel indicators:

- **Decoder Net Gain ($G(v) = C_f(v) - C_e(v) = P(v) - N(v)$):** Measures the net directional balance ($G(v) = +1$ for correction, -1 for degradation, 0 for unchanged).
- **Correctness Activity ($A_{\text{corr}}(v) = P(v) + N(v) = |G(v)|$):** Measures where additional decoder capacity alters prediction correctness (gross activity).

3.3. Global Rates and Evaluation Domains

To aggregate voxel-level indicators into volume-level rates for subject i , we define the positive correction rate $R_+^{(i)}$, negative degradation rate $R_-^{(i)}$, gross activity rate $R_{\text{act}}^{(i)}$, and net gain rate $R_{\text{net}}^{(i)}$ over an evaluation domain Ω_i :

$$R_+^{(i)} = \frac{1}{|\Omega_i|} \sum_{v \in \Omega_i} P_i(v), \quad R_-^{(i)} = \frac{1}{|\Omega_i|} \sum_{v \in \Omega_i} N_i(v) \quad (1)$$

$$R_{\text{act}}^{(i)} = R_+^{(i)} + R_-^{(i)}, \quad R_{\text{net}}^{(i)} = R_+^{(i)} - R_-^{(i)} \quad (2)$$

We evaluate these rates across two complementary spatial domains Ω_i :

1. **All-Volume Domain (Ω_{all}):** Encompasses all voxels within the 3D volume grid V_i , yielding $R_{\text{act}}^{\text{all}}$ and $R_{\text{net}}^{\text{all}}$. This domain measures aggregate computational efficiency over the complete volumetric field.
2. **Foreground-Union Domain (Ω_{fg}):** Restricts evaluation to voxels belonging to ground-truth or predicted foreground structures ($\Omega_{\text{fg}} = \{v \in V_i \mid y_i(v) > 0 \vee \hat{y}_{e,i}(v) > 0 \vee \hat{y}_{f,i}(v) > 0\}$), yielding $R_{\text{act}}^{\text{fg}}$ and $R_{\text{net}}^{\text{fg}}$. By filtering out uninformative background voxels, this domain isolates localized prediction shifts directly on anatomical target regions.

Global metrics are reported as population means with 95% confidence intervals computed via 1000-resample paired bootstrapping across subjects, preserving subject-level correlation.

3.4. Characterization Metrics and Evaluation Protocol

We structure our characterization metrics along three primary evaluation axes: spatial anatomical heterogeneity, positive-gain concentration, and inference-time predictability (Table 1).

3.4.1. Anatomical Boundary Stratification

To analyze how correctness changes distribute relative to anatomical structures, we measure distance strictly to the **ground-truth anatomical boundary** (avoiding circularity from model-predicted boundaries) in physical millimeters:

$$d_{GT}(v) = \min_{u \in \partial S_{GT}} \|\mathbf{D}(x_v - x_u)\|_2, \quad (3)$$

where $\mathbf{D} = \text{diag}(s_x, s_y, s_z)$ incorporates dataset physical voxel spacing. We stratify correctness activity $A_{\text{corr}}(v)$ and net gain

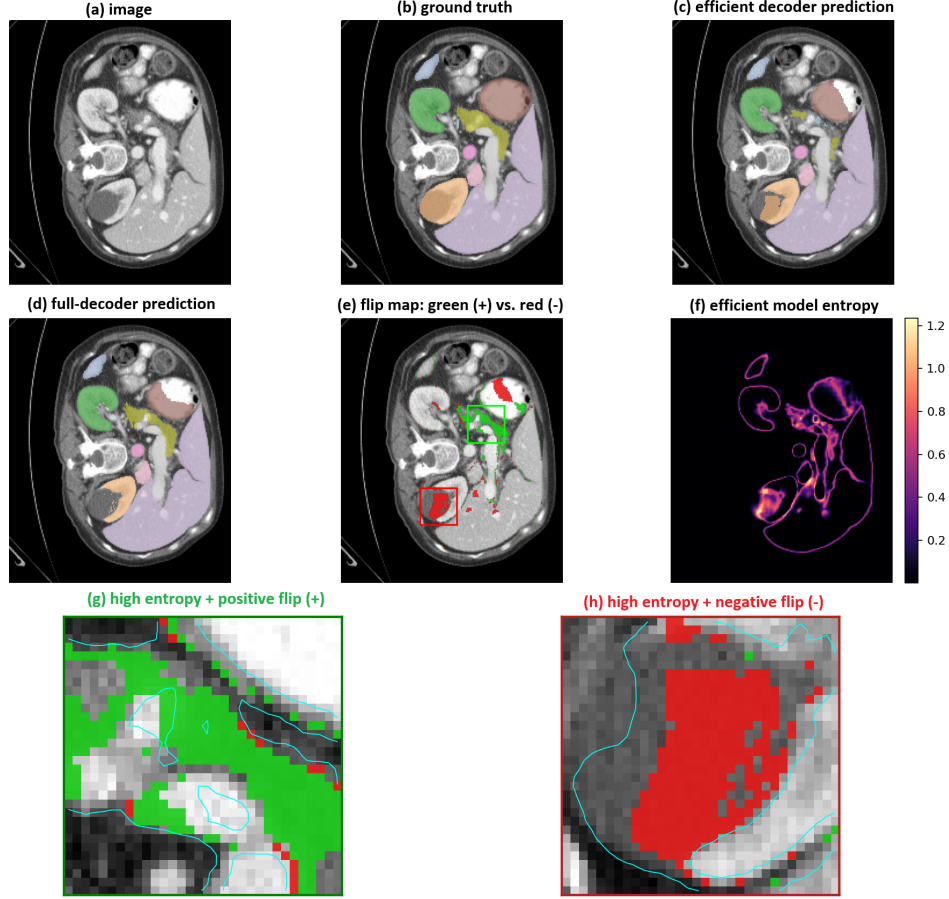


Figure 2: Qualitative illustration of correctness activity and directional gain for SwinUNETR on BTCV CT dataset under Full (E_0) vs. Efficient (E_1) decoders (see Supplementary Figures S1 and S2 for 3D UX-Net and MedNeXt-M-K3).

Table 1: Overview of analysis objectives, evaluation metrics, scientific focus, and section references.

Analysis Objective	Evaluation Metrics	Scientific Focus	Section & Eq. Ref.
Heterogeneity			
Gross Voxel Shift	Correctness Activity $A_{\text{corr}}(v)$	Identify spatial sites of decoder correctness shifts	Sec. 3.2
Directional Effect	Net Gain $G(v)$	Quantify directional balance of corrections vs. degradations	Sec. 3.2
Global Rates	Subject Rates ($R_{\text{act}}, R_{\text{net}}$)	Measure population rates over all-volume and foreground domains	Sec. 3.3, Eq. (1)–(2)
Boundary Stratification	Distance Bands (d_{GT})	Stratify correctness activity by physical boundary distance	Sec. 3.4.1, Eq. (3)
Concentration			
Oracle Benefit Curve	Oracle Coverage Curve	Track cumulative positive gain across spatial blocks	Sec. 3.4.3, Eq. (4)
Concentration Index	Oracle Concentration K_{80}	Determine minimum volume fraction carrying 80% gain	Sec. 3.4.3
Predictability			
Activity Localization	Location AUROC	Evaluate inference-time signal localization of active blocks	Sec. 3.4.4
Utility Discrimination	Direction AUROC	Assess signal discrimination of beneficial vs. harmful flips	Sec. 3.4.4
Multi-View Disagreement	TTA Mutual Info $I(v)$	Test epistemic disagreement for direction predictability	Sec. 3.4.4
Retrospective Benefit	Hybrid Dice Recovery $R_{\text{rec}}(k)$	Upper-bound recoverable macro-Dice under spatial budget k	Sec. 3.4.5, Eq. (5)

$G(v)$ across physical distance bands (0–2, 2–4, 4–8, and > 8 mm) to isolate partial-volume boundary interfaces from homogeneous organ interiors.

MSD08: $30 \times 30 \times 16$ voxels). A block b_r is active if $A_{\text{corr}}(r) = \mathbf{1}[\sum_{v \in b_r} A_{\text{corr}}(v) > 0]$, and its directional gain is $G(r) = \sum_{v \in b_r} G(v)$.

3.4.2. Spatial Block Partitioning

To handle spatial sparsity and noise, voxels are aggregated into contiguous cubic spatial blocks covering approximately $24 \times 24 \times 24 \text{ mm}^3$ (BTCV: $16 \times 16 \times 12$; MSD01: $24 \times 24 \times 24$;

3.4.3. Gain Concentration (K_{80})

Sorting spatial blocks by descending net gain $G(r)$, the oracle coverage curve accumulated over positive net gain is:

$$\text{Coverage}(k) = \frac{\sum_{r=1}^{\lfloor (k/100) \cdot |B| \rfloor} \max(G(r), 0)}{\sum_{r=1}^{|B|} \max(G(r), 0)}. \quad (4)$$

We report K_{80} , defined as the minimum volume fraction of spatial blocks an oracle selector must retain to capture 80% of all positive net gain [22]. Lower K_{80} values denote higher spatial concentration of useful decoding work.

3.4.4. Predictability Scores

To test whether cheap inference-time signals anticipate decoder utility, we evaluate single-pass Shannon entropy $H(v) = -\sum_c p_c(v) \log p_c(v)$ [23, 15, 24] and 8-view Test-Time Augmentation Mutual Information (TTA-MI) $I(v)$. We define two distinct evaluation tasks: **Location Predictability** (predicting active blocks $A_{\text{corr}}(r) > 0$) and **Direction Predictability** (distinguishing beneficial $G(r) > 0$ from harmful $G(r) < 0$ blocks) [21].

3.4.5. Retrospective Dice Recovery (R_{rec})

Given a selected block set S_k , we form a retrospective hybrid prediction $\hat{y}_k(v)$ that substitutes full decoder predictions inside S_k while keeping efficient baseline predictions elsewhere. The fraction of recovered macro-Dice gap is:

$$R_{\text{rec}}(k) = \frac{\text{Dice}(\hat{y}_k) - \text{Dice}(\hat{y}_e)}{\text{Dice}(\hat{y}_f) - \text{Dice}(\hat{y}_e)}. \quad (5)$$

Evaluating $R_{\text{rec}}(k)$ across varying spatial budgets k establishes an empirical upper bound on recoverable segmentation quality under spatial compute constraints.

3.4.6. Factorial Factor Attribution

Under a frozen shared encoder, we evaluate a 2×2 factorial grid across channel width $C \in \{C_{384}, C_{48}\}$ and resolution scaling $RF \in \{RF_1, RF_2\}$. We compute channel main effect Δ_C , resolution main effect Δ_{RF} , and interaction $\Delta_{C \times RF}$ to isolate the architectural drivers of decoder performance.

4. Experimental Setup

This section describes the target 3D backbones, clinical datasets, evaluation axes, and implementation setup. Our design triangulates decoder computation with three complementary probes on a shared 3D UX-Net / BTCV anchor (Figure 3): an ecological *generality* axis (matched full/efficient pairs across backbones and datasets), a *causal* frozen shared-encoder 2×2 factorial, and a *logic-robustness* control that replaces EffiDec3D’s capacity removal with a capacity-preserving depthwise-separable decoder. The three probes differ in training regime (end-to-end vs. frozen encoder) and efficiency mechanism (capacity removal vs. preservation), so agreement among them is not built into any single intervention form.

4.1. Target Architectures and Intervention Specifications

We evaluate matched decoder pairs across three representative 3D medical image segmentation backbones covering distinct architectural paradigms: a large-kernel CNN (3D UX-Net [5]), a vision transformer (SwinUNETR [4]), and a ConvNeXt-style network (MedNeXt-M-K3 [6]). Interventions occur strictly within the decoder pathway (upsampling, skip fusion, and refinement) while preserving identical encoder weights and feature dimensions. For each backbone, the efficient decoder baseline (E_1) implements a standardized capacity reduction: an 87.5% feature channel compression ($C_{\text{full}} \rightarrow C_{\text{reduced}}$) and the omission of the highest-resolution skip-refinement stage (Stage 1), following the architecture-preserving principles of EffiDec3D [7]. All parameters, FLOPs (profiled on a standardized $1 \times C \times 96 \times 96 \times 96$ patch), and volumetric inference latencies are summarized in Table 2.

As a second, structurally different intervention template, we additionally train a *depthwise-separable* decoder on the 3D UX-Net anchor. Unlike EffiDec3D, it removes no capacity: every channel width and high-resolution stage of the full decoder (E_0) is retained, and only each dense $3 \times 3 \times 3$ decoder convolution is factorized into a depthwise plus a pointwise convolution (reducing a dense $192 \rightarrow 192$ conv from 995K to 42K parameters). This lightens the decoder through an efficiency mechanism orthogonal to capacity pruning, and serves as the logic-robustness control of Section 5.6.

4.2. Factorial Configurations

To independently attribute decoder contributions to channel capacity versus spatial resolution scaling, we construct a 2×2 factorial grid on the 3D UX-Net backbone under a frozen, shared encoder. The grid crosses two channel capacity levels (C_{384} vs. C_{48}) with two high-resolution stage retention levels (RF_1 : retain full-resolution stage vs. RF_2 : omit full-resolution stage). Note that while the end-to-end reference baseline (Table 2) incorporates auxiliary deep-supervision heads (578.74 GMac), the standalone factorial decoder isolates the decoding pathway with auxiliary heads detached (657.76 GMac under standalone patch profiling), enabling exact factor attribution without confounding loss terms.

4.3. Clinical Datasets and Characteristics

To evaluate decoder computation across diverse anatomical targets, clinical tasks, and imaging modalities, we conduct experiments on three representative 3D volumetric benchmarks:

1. **BTCV Abdominal Multi-Organ CT Benchmark** [25]: Captures large-scale multi-class organ segmentation across 13 abdominal structures (30 CT volumes). It features diverse anatomical scales—ranging from large homogeneous organs (liver, spleen) to small, highly variable structures (adrenal glands, gallbladder)—with well-defined organ boundaries and organ-to-background interfaces.
2. **MSD01 Brain Tumour Multi-Modal MRI Benchmark** [26, 27]: Captures neuro-oncological tissue segmentation across 484 multi-sequence MRI volumes (FLAIR,

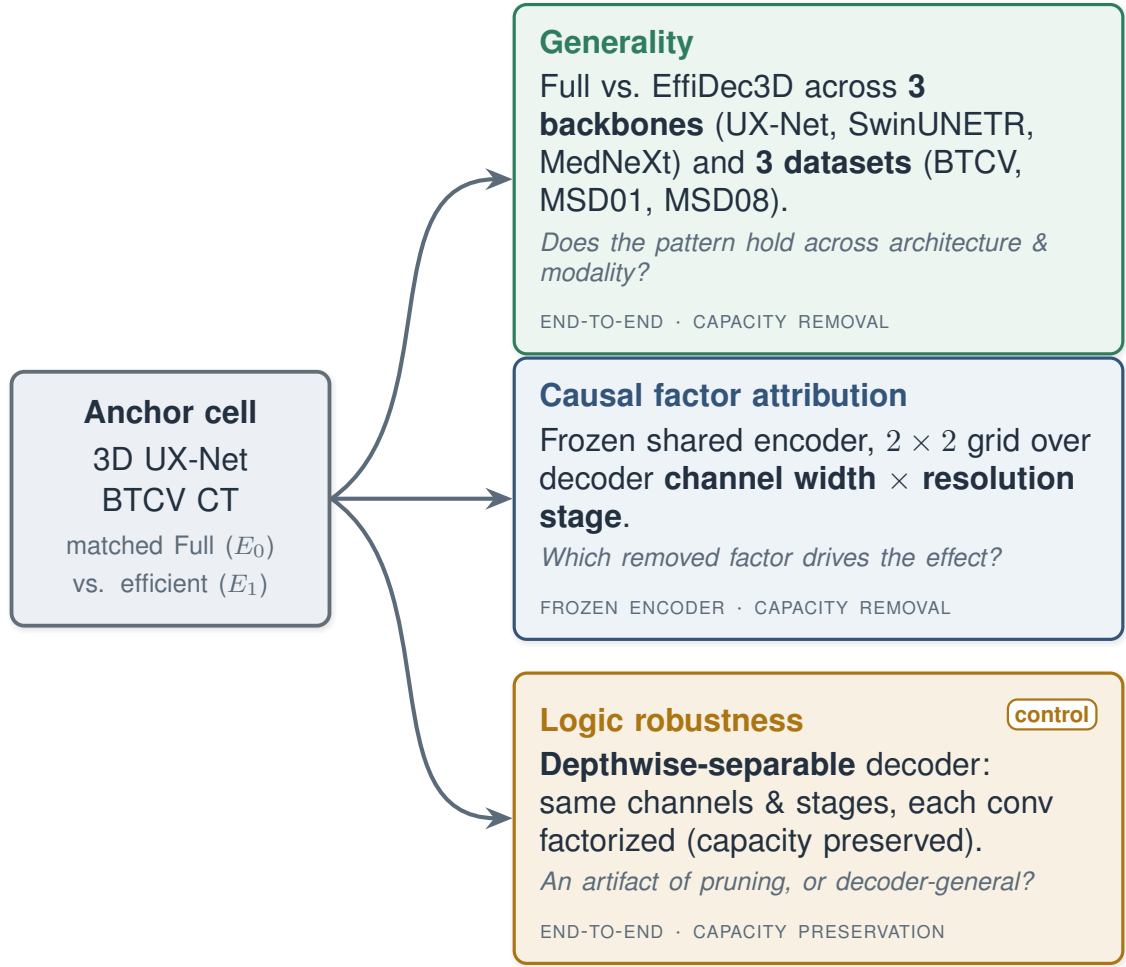


Figure 3: Experimental design: three complementary probes of decoder computation on a shared 3D UX-Net / BTCV anchor. They differ in training regime (end-to-end vs. frozen shared encoder) and efficiency mechanism (capacity removal vs. preservation), jointly testing generality across backbones and datasets, causal attribution of the removed factors, and robustness to the decoder-lightening logic.

Table 2: Comprehensive quantitative comparison of Full (E_0) vs. Efficient (E_1) decoders across three 3D backbones on the BTCV 13-organ CT dataset, detailing decoder architectural configurations, computational complexity, inference latency, and segmentation performance.

Architecture	Decoder Variant	Decoder Config	Params (M) ↓	FLOPs (GMac) ↓	Latency (ms) ↓	Mean Dice (%) ↑	Mean HD95 (mm) ↓
3D UX-Net [5]	Full (E_0)	C_{384} , $1 \times$ resolution	53.01	578.74	40.6	79.18	9.04
	Effi (E_1)	C_{48} , omit Stage 1 (−87.5%)	3.16	49.49	8.0	77.73	11.82
SwinUNETR [4]	Full (E_0)	C_{384} , $1 \times$ resolution	62.19	308.24	37.0	80.48	8.88
	Effi (E_1)	C_{48} , omit Stage 1 (−87.5%)	11.21	55.84	16.7	79.49	8.90
MedNeXt-M-K3 [6]	Full (E_0)	C_{256} , $1 \times$ resolution	17.56	106.34	27.8	82.61	7.25
	Efficient (E_1)	C_{32} , omit Stage 1 (−87.5%)	5.80	49.54	12.9	81.35	6.77

T1, T1ce, T2). It encompasses three sub-regions (edema, non-enhancing core, enhancing tumor) characterized by diffuse, infiltrative margins and highly irregular spatial topologies.

- MSD08 Hepatic Vessel CT Benchmark** [27]: Captures fine-grained abdominal vascular segmentation across 303 CT volumes. It focuses on thin, tubular vessel branches and liver tumors, presenting extreme foreground sparsity,

complex branching geometry, and high surface-area-to-volume ratios.

Target voxel spacings ($1.5 \times 1.5 \times 2.0 \text{ mm}^3$ for BTCV, $1.0 \times 1.0 \times 1.0 \text{ mm}^3$ for MSD01, and $0.8 \times 0.8 \times 1.5 \text{ mm}^3$ for MSD08) and physical spatial block partitioning follow the standardized protocol established in Section 3.4.2 to maintain an invariant physical block footprint of approximately $24 \times 24 \times 24 \text{ mm}^3$ across all modalities.

4.4. Evaluation Strategy and Experimental Axes

To systematically map decoder contributions without evaluating an unconstrained, dense Cartesian product across all models and datasets, we organize our evaluation along three targeted experimental axes:

1. **Cross-Architecture Consistency Axis:** Evaluates matched full (E_0) vs. efficient (E_1) decoders across three distinct backbones (3D UX-Net, SwinUNETR, MedNeXt-M-K3) on the canonical BTCV CT dataset to establish architectural consistency across model families.
2. **Cross-Dataset Consistency Axis:** Evaluates matched decoder pairs using a fixed reference backbone (3D UX-Net) across three clinical benchmarks (BTCV CT, MSD01 Brain MRI, MSD08 Hepatic CT) to test generalization across imaging modalities and anatomical topologies.
3. **Decoder-Isolated Factor Attribution Axis:** Deconstructs decoder capacity into a 2×2 factorial grid (C_{384} vs. C_{48} , RF_1 vs. RF_2) under a frozen shared 3D UX-Net encoder on BTCV CT, strictly isolating channel capacity from resolution scaling without confounding encoder weight shifts.

4.5. Implementation and Statistical Protocol

All models were implemented in PyTorch 2.6 with CUDA 12.8 using an NVIDIA RTX 5090 GPU. Training was conducted for 45,000 optimizer iterations using bfloat16 mixed precision, the AdamW optimizer (initial learning rate 1×10^{-3} with cosine annealing, weight decay 1×10^{-5}), and a compound Dice-CrossEntropy loss function on $96 \times 96 \times 96$ voxel crops under standard 3D spatial augmentations. To guarantee exact paired comparisons, matched full and efficient decoders were trained and evaluated on identical subject-level volume splits. Model checkpoints were selected strictly by validation macro-Dice, and characterization metrics were evaluated on held-out test subjects, reporting population means with subject-level paired bootstrap 95% confidence intervals (1000 resamples).

5. Results and Discussion

In this section, we present empirical characterization findings across target backbones, clinical datasets, and intervention designs, synthesizing numerical findings directly with their architectural, statistical, and clinical implications.

5.1. Macro Trade-offs and Bidirectional Voxel Dynamics

Evaluating aggregate segmentation accuracy alongside computational complexity reveals a well-known engineering phenomenon: decoder simplification yields massive computational savings at minimal cost to macro-level Dice scores. As detailed in Table 2, replacing the full decoder (E_0) with an efficient baseline (E_1) removes up to 91.4% of FLOPs and reduces inference latency by up to $5\times$ (3D UX-Net: 40.6 ms $\rightarrow$ 8.0 ms; SwinUNETR: 37.0 ms $\rightarrow$ 16.7 ms; MedNeXt-M-K3: 27.8 ms $\rightarrow$ 12.9 ms). Across all three backbones, the corresponding Mean Dice penalty remains within 1.45 percentage

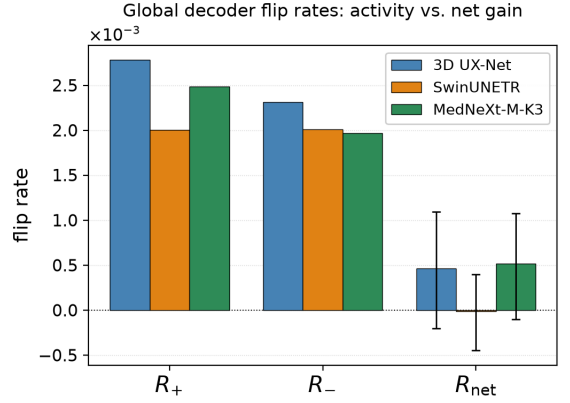


Figure 4: Global flip rates (R_+ , N , R_{net}) across backbones under Full (E_0) vs. Efficient (E_1) decoders on BTCV CT.

points (79.18% $\rightarrow$ 77.73% for 3D UX-Net, 80.48% $\rightarrow$ 79.49% for SwinUNETR, and 82.61% $\rightarrow$ 81.35% for MedNeXt-M-K3), while boundary error (HD95) even shows slight numerical improvements for two of the three backbones.

Read at face value, these aggregate numbers appear to justify aggressive decoder pruning as a near-lossless optimization. However, a macro-averaged Dice score is a single scalar pooled over millions of voxels. Isolating voxel-level prediction shifts between matched decoder pairs exposes a fundamentally different underlying reality: macro-level metric stability masks extensive voxel-level prediction reorganization.

Evaluating voxel predictions demonstrates that decoder activity is governed by strong bidirectional cancellation ($P(v) \approx N(v)$), rendering aggregate net volume-wide gain statistically near-zero (Figure 4). Positive voxel corrections $P(v)$ (where E_0 fixes an E_1 error) are substantially counterbalanced by negative degradations $N(v)$ (where E_0 degrades a correct E_1 prediction). Over the All-Volume domain (Ω_{all}), subject-mean net gain rates R_{net}^{all} remain minimal: +0.046% (95% CI: [-0.020%, +0.110%]) for 3D UX-Net, -0.001% (95% CI: [-0.045%, +0.040%]) for SwinUNETR, and +0.052% (95% CI: [-0.010%, +0.108%]) for MedNeXt-M-K3. Even over the Foreground-Union domain (Ω_{fg}), net gain rates stay small (+0.381% for 3D UX-Net, -0.008% for SwinUNETR, +0.446% for MedNeXt-M-K3), and all paired subject-bootstrap 95% confidence intervals include zero. Rather than acting as a monotonic error corrector, high-capacity decoders perform spatial prediction reallocation whose net volume-wide effect is near-neutral.

5.2. Spatial and Anatomical Heterogeneity

The near-zero aggregate net gain is not the signature of an inert decoder, but of intense activity localized to specific spatial zones and anatomical structures.

Stratifying voxel correctness activity $A_{corr}(v)$ by physical Euclidean distance $d_{GT}(v)$ to the true ground-truth boundary (Figure 5) shows a sharp peak within the innermost 0–2 mm boundary band, followed by rapid decay across 2–4 mm and 4–8 mm, reaching virtual inactivity in homogeneous organ interiors ($d_{GT} > 8$ mm). Additional decoder computation is con-

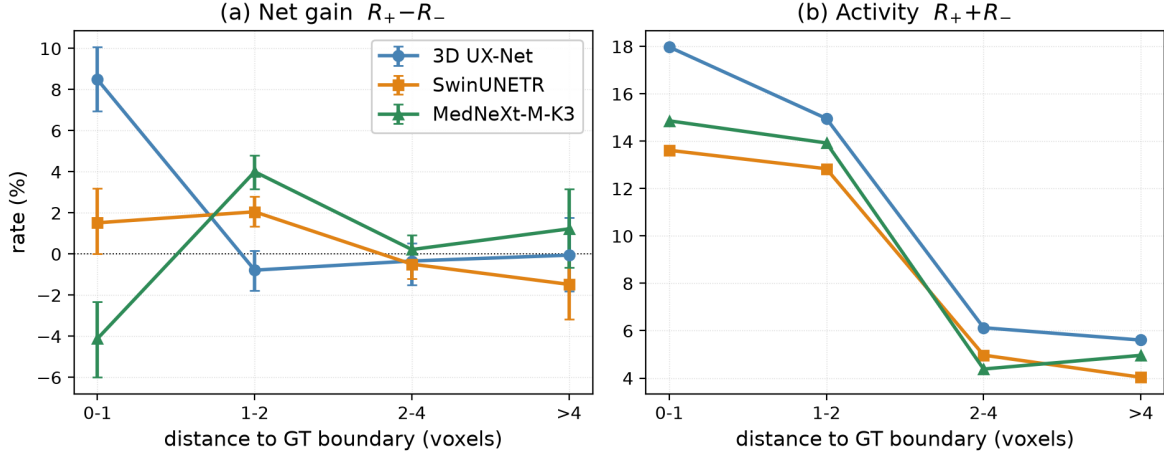


Figure 5: Boundary-resolved decoder benefit versus physical distance to ground-truth boundary (d_{GT}) across backbones: (a) net gain $R_+ - R_-$, and (b) activity $R_+ + R_-$.

summed almost exclusively in partial-volume transition zones where boundary ambiguity is highest. This spatial concentration confirms that feature representations within organ interiors are already well-resolved by the encoder, rendering high-resolution decoding pathways in interior regions computationally redundant.

This spatial pattern translates directly into an anatomical scale effect across organs (Figure 6). Small, morphologically complex structures with high surface-to-volume ratios exhibit the highest activity rates: left and right adrenal glands reach positive correction rates up to 16.9% in 3D UX-Net (14.8% in SwinUNETR, 14.0% in MedNeXt-M-K3), alongside high correction rates for the gallbladder and esophagus. Conversely, large compact organs such as the liver, spleen, and kidneys remain nearly inert (<3% activity) because their volume is dominated by interior voxels rather than boundary margins. Thus, anatomical scale directly dictates decoder workload.

The Metric Weighting Paradox.. This anatomical separation resolves a key conceptual paradox: *How can additional decoder computation improve macro-Dice while producing near-zero global voxel net gain?* Global voxel correctness is mathematically dominated by vast background regions and large organ interiors, where predictions are invariant to decoder capacity. In contrast, macro-averaged Dice weights all anatomical classes equally and is foreground-sensitive. Positive corrections concentrated on small, volatile structures (e.g., adrenal glands) exert a disproportionate impact on macro-Dice, even while global voxel shifts remain net-neutral across the full volumetric grid.

5.3. Architectural Drivers and Spatial Concentration

To disentangle the architectural drivers of these spatial shifts, Table 3 and Table 4 report our frozen-encoder 2×2 factorial analysis on 3D UX-Net, isolating resolution scaling (RF) from feature channel capacity (C_{dec}). Holding the encoder strictly constant ensures that observed prediction shifts stem exclusively from decoder interventions rather than encoder weight adaptation.

High-resolution stage retention ($RF_1 \rightarrow RF_2$) accounts for a major FLOP reduction ($657.8 \rightarrow 319.1$ GMac, $\Delta_{RF} = -338.66$ GMac) and controls spatial boundary refinement (+0.033% net gain in 0–2 mm band, decaying to -0.011% in interiors). Channel width reduction ($C_{384} \rightarrow C_{48}$) cuts parameter count by 95% ($46.7M \rightarrow 2.2M$) while exerting a spatially diffuse, net-neutral effect ($\Delta_C = -0.020\%$). The near-zero interaction term ($\Delta_{C \times RF} = -0.004\%$) indicates that channel width and resolution scaling act approximately additively, allowing them to be decoupled in efficient decoder design.

Sorting spatial blocks by descending net gain (Figure 7) reveals that an oracle selector retaining just the top 5% of spatial blocks captures at least 80% of all positive net gain ($K_{80} \leq 5\%$) across all three backbones, with the top 10% capturing up to 99.98%. This confirms that the vast majority of decoder computation is spatially redundant, establishing the empirical precondition for budgeted, dynamic spatial decoding.

5.4. Inference-Time Predictability and Selective Recovery

While oracle concentration establishes that a small volume fraction *could* carry decoder benefit, practical execution requires anticipating that benefit at inference time using cheap baseline signals.

As shown in Table 5, single-pass predictive Shannon entropy achieves high Location AUROC (≥ 0.900 – 0.913), reliably detecting active blocks. However, Direction AUROC drops to near-chance levels (0.556 – 0.591). Multi-sample epistemic disagreement via Test-Time Augmentation Mutual Information ($I(v)$) similarly yields modest direction AUROCs (0.539 – 0.570). Figure 8 exposes the physical cause: positive corrections $P(v)$ and negative degradations $N(v)$ share nearly identical conditional entropy distributions $p(H | P(v) = 1) \approx p(H | N(v) = 1)$, rendering them directionally indistinguishable from uncertainty alone. This exposes the fundamental *Activity–Direction Predictability Gap*: predictive uncertainty identifies where computation alters predictions, but fails to predict whether those alterations will be beneficial or harmful.

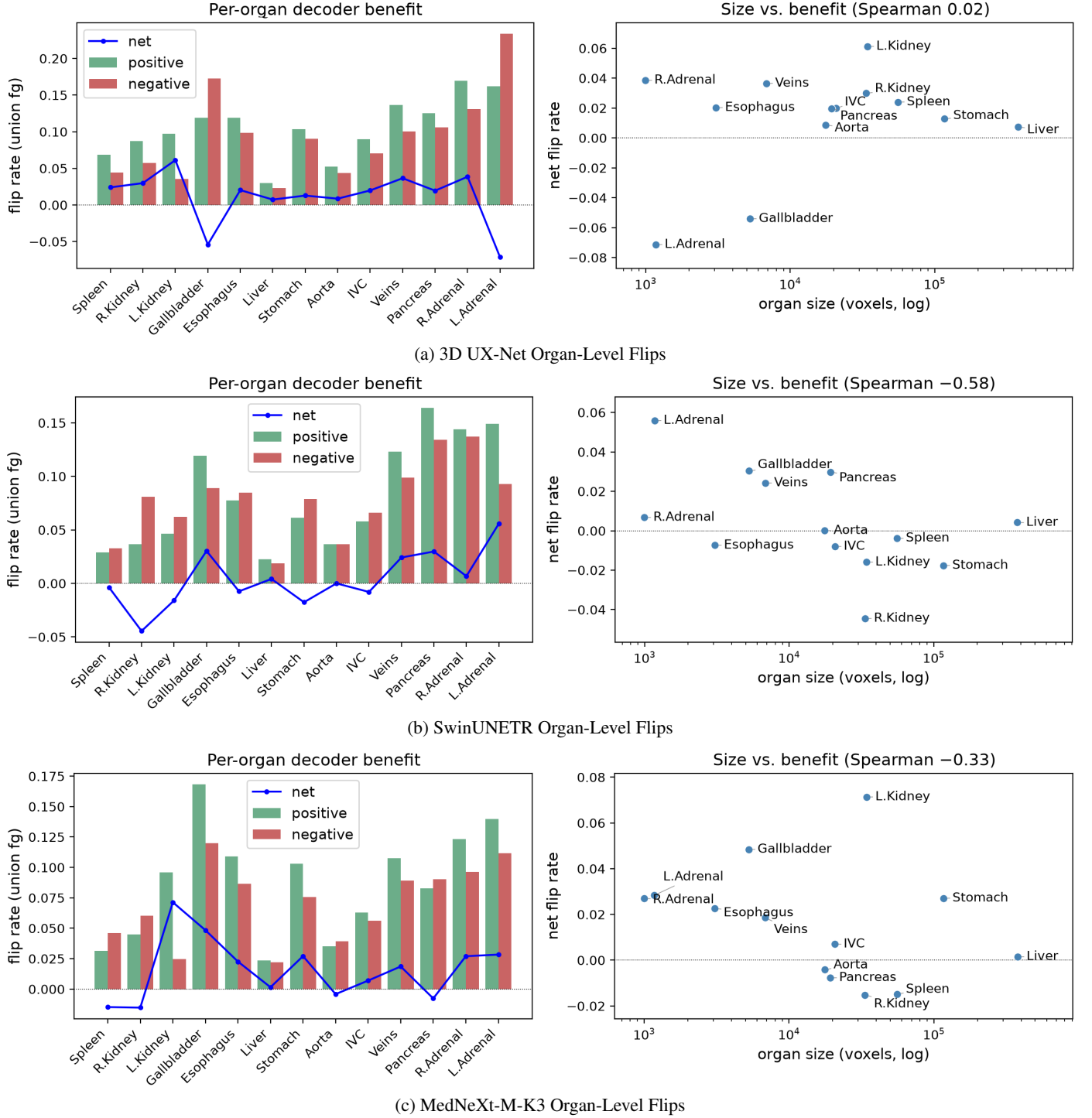


Figure 6: Organ-level flip breakdown across backbones under Full (E_0) vs. Efficient (E_1) decoders (BTCV CT).

Despite direction ambiguity, uncertainty locates activity well enough to guide spatial allocation. Under retrospective output-space hybridization (Figure 9), substituting full decoder predictions in high-entropy blocks recovers 60% of the full-volume Dice gap at a 5% spatial budget, and achieves $\geq 100\%$ full gap recovery at a 20% spatial budget across all backbones (versus only 17–20% for random block selection). Crucially, because high-entropy spatial blocks concentrate the vast majority

of gross correctness activity (A_{corr}), routing the full decoder to these candidate patches ensures that positive corrections $P(v)$ on volatile structures are captured within the retained 20% spatial budget. This demonstrates that selectively routing computation to high-uncertainty regions can recover macroscopic performance while skipping heavy decoding across 80% of the volumetric grid.

Table 3: 2×2 Factorial analysis of decoder channel reduction (C_{dec}) and resolution scaling (RF) on 3D UX-Net and BTCV CT (Decoder-Isolated Factor Attribution Axis).

Configuration	C_{dec}	RF	Params (M)	FLOPs (GMac)	Latency (ms)	Mean Dice (%)
Full (C_{384}, RF_1)	384	1	46.72	657.76	43.3	80.44
Chan-Only (C_{48}, RF_1)	48	1	2.24	388.15	35.2	79.80
Res-Only (C_{384}, RF_2)	384	2	46.32	319.10	16.1	77.97
Effi (C_{48}, RF_2)	48	2	1.84	49.49	8.0	78.35

Table 4: Factor main effects (Δ_C, Δ_{RF}) and interaction ($\Delta_{C \times RF}$) across metrics on 3D UX-Net (BTCV CT).

Outcome Metric	Channel Δ_C	Stage Δ_{RF}	Interaction $\Delta_{C \times RF}$
Mean Dice (%)	-0.13	-1.96	+1.02
Params (M)	-44.48	-0.40	+0.00
FLOPs (GMac)	-269.61	-338.66	+0.00
Boundary Net Gain (%)	-0.020	+0.033	-0.004

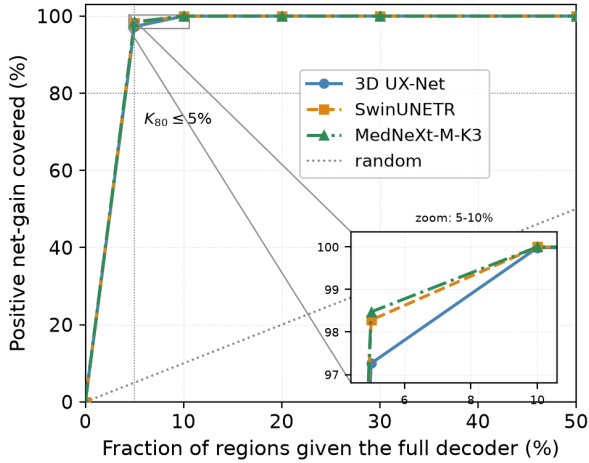


Figure 7: Oracle positive-gain concentration curves (K_{80}) across backbones.

Table 5: Inference-time predictability of block-level activity (Location AUROC) vs. directional utility (Direction AUROC) across backbones.

Backbone	Location AUROC	Direction AUROC
3D UX-Net	0.900	0.591
SwinUNETR	0.913	0.560
MedNeXt-M-K3	0.907	0.556

5.5. Cross-Dataset Generalization

Table 6 validates these characterization findings across three clinical imaging datasets using the 3D UX-Net reference architecture.

High location AUROC (0.865–0.900), low direction AUROC (0.520–0.591), and extreme benefit concentration ($K_{80} \leq 5\%$) persist consistently across multi-organ CT, multi-sequence brain MRI, and hepatic vascular CT. However, the sign and magnitude of net directional gain are task-dependent. On MSD08 HepaticVessel CT, the efficient decoder slightly outperforms the full decoder (58.63% vs. 57.51% Dice; $R_{net}^{fg} = -4.89\%$). Because fine vascular structures possess extreme

surface-to-volume ratios, additional decoder capacity in high-resolution stages can overfit ambiguous boundary noise. This critical finding proves that extra decoder computation is not monotonically beneficial, reinforcing the necessity of utility-gated dynamic computation.

5.6. Robustness to the Decoder-Lightening Mechanism

All results so far obtain the efficient decoder by EffiDec3D’s capacity removal (channel reduction plus high-resolution stage omission). To test whether the characterization is an artifact of that specific pruning rather than a property of decoder computation, we repeat the anchor-cell analysis with a structurally different, capacity-preserving efficient decoder: a depthwise-separable decoder that keeps every channel and stage of the full decoder and only factorizes each convolution (Section 4.1). Table 7 compares the two efficient decoders on 3D UX-Net / BTCV, with every metric computed identically.

Each qualitative finding replicates under the separable decoder: bidirectional activity (0.50%) far exceeds the net gain; the positive net that exists is concentrated at the immediate ground-truth boundary (+1.58% in the 0–1 vox band); location predictability far exceeds direction predictability (0.886 vs. 0.582); oracle concentration is near-total (99.4% of positive gain within the top 5% of blocks, $K_{80} \leq 5\%$); and a 20% entropy budget recovers the full macro-Dice gap. The values track the EffiDec3D pair closely despite an orthogonal efficiency mechanism.

One difference is worth stating plainly. The separable decoder agrees with the full decoder far more (only 0.50% activity), and its net gain, though tiny, is marginally positive (+0.068%, 95% CI [+0.010, +0.133]) rather than strictly zero-crossing as for the removal-based pairs. That residual benefit is itself boundary-localized, so it sharpens rather than contradicts the central picture: a capacity-preserving decoder leaves the full model only a hair of genuine benefit, and that hair lives at the boundary. Both efficiency mechanisms therefore yield the same characterization—highly active, near-neutral, boundary-localized, direction-unpredictable, concentrated, and recoverable—so the findings are not tied to *how* the decoder is made efficient.

5.7. Clinical Stratification and Decoder Design Principles

Translating these characterization findings into clinical practice yields a clear anatomical stratification for decoder deployment. Large, compact organs (e.g., liver, spleen, kidneys) are dominated by interior voxels and exhibit minimal activity (<3%), meaning efficient decoders can be safely deployed to

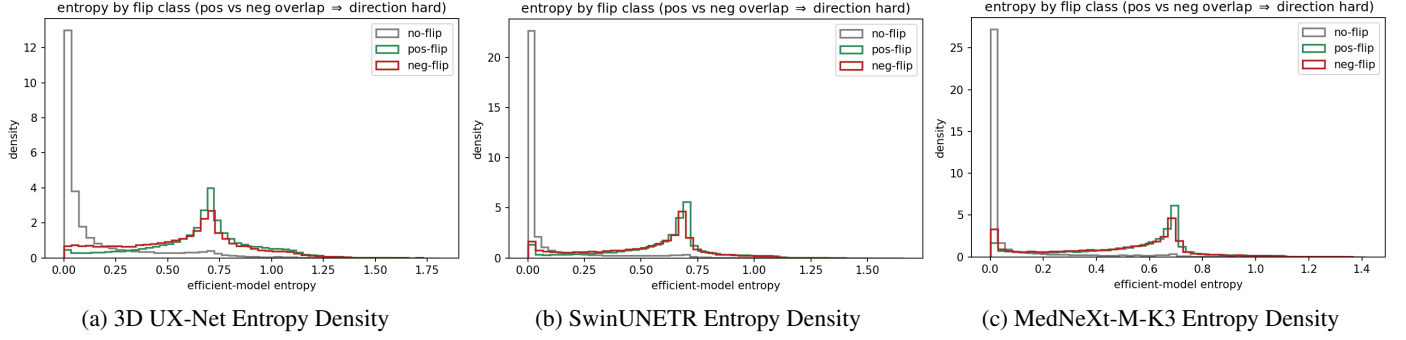


Figure 8: Conditional predictive entropy density distributions $p(H | P = 1)$ vs. $p(H | N = 1)$ across backbones.

Table 6: Cross-dataset generalization validation using 3D UX-Net across three clinical imaging datasets (Cross-Dataset Consistency Axis).

Dataset	Full Dice (%)	Effi Dice (%)	Activity R_{act}^{all} (%)	Net Gain R_{net}^{all} (%)	Activity R_{act}^{fg} (%)	Net Gain R_{net}^{fg} (%)	Oracle K_{80}	Location AUROC	Direction AUROC
BTCV CT	79.18	77.73	3.42	+0.046	12.84	+0.381	$\leq 5\%$	0.900	0.591
MSD01 BrainTumour	75.55	74.95	0.46	+0.038	18.10	+2.86	$\leq 5\%$	0.869	0.558
MSD08 HepaticVessel	57.51	58.63	0.20	-0.002	22.99	-4.89	$\leq 5\%$	0.865	0.520

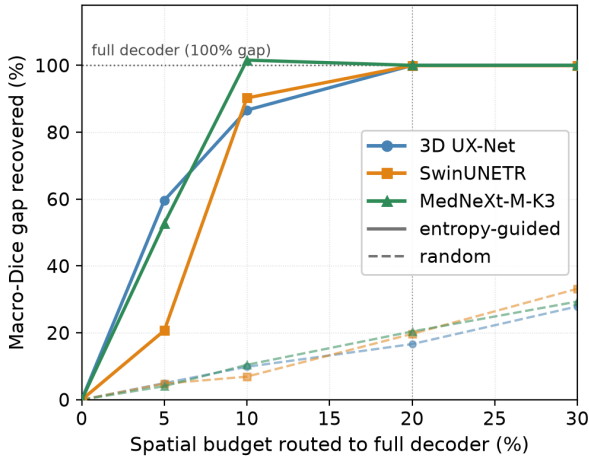


Figure 9: Macro-Dice gap recovery trajectories under retrospective output hybridization across backbones.

maximize inference throughput in high-throughput screening or intraoperative guidance workflows. Conversely, small, morphologically complex structures (e.g., adrenal glands, gallbladder, vessels) exhibit high boundary activity (up to 16.9%), requiring high-resolution decoder capacity or targeted entropy-guided refinement for precise resection planning and vessel delineation. Finally, on thin vascular networks (e.g., MSD08), additional decoder capacity can overfit boundary noise, proving that computation should be gated based on expected gain rather than applied uniformly by default.

These empirical insights establish three practical principles for future dynamic 3D decoders:

1. **Separate activity localization from utility estimation:** Predictive uncertainty locates candidate regions effectively but cannot predict gain direction; routing policies must pair location proposals with directional gain estimators to

Table 7: Robustness of the characterization to the decoder-lightening mechanism on 3D UX-Net / BTCV: EffiDec3D capacity removal vs. a capacity-preserving depthwise-separable decoder. All metrics are computed identically for both pairs.

Metric (UX-Net / BTCV)	EffiDec3D (removal)	Separable (factorized)
Full / efficient Dice	0.791 / 0.776	0.799 / 0.785
Activity $R_+ + R_-$ (%)	0.51	0.50
Net gain $R_+ - R_-$ (%)	+0.047	+0.068
95% CI	$[-0.020, +0.109]$	$[+0.010, +0.133]$
Boundary net, 0–1 vox (%)	+8.49	+1.58
Location AUROC	0.900	0.886
Direction AUROC	0.591	0.582
Oracle coverage @5%	97.3%	99.4%
Entropy recovery @20%	100%	100%

prevent harmful degradations.

2. **Decouple globally shared decoding from local high-resolution refinement:** Channel capacity can be compressed globally with minimal spatial impact, whereas high-resolution stages should be selectively activated only on ambiguous boundary patches.
3. **Treat additional computation as a task-dependent intervention:** Dynamic execution should default to a lightweight base decoder, allocating heavy high-resolution refinement only when the expected directional gain exceeds computational execution overhead.

5.8. Limitations

While this study provides a comprehensive characterization of 3D decoder computation, three primary limitations should be noted:

First, our controlled interventions draw from architecture-preserving channel compression and stage omission templates

(inspired by EffiDec3D and depthwise-separated decoder designs). Although validated across diverse CNN, ViT, and ConvNeXt backbones, future work may explore dynamic gating or non-uniform depth pruning to examine whether unconstrained architecture search yields different spatial redundancy patterns.

Second, selective spatial recovery is established via retrospective output-space hybridization. This defines an empirical opportunity bound for localized allocation rather than demonstrating an integrated, hardware-level dynamic execution engine with real-time runtime speedup and CUDA kernel acceleration.

Third, inference-time predictability was evaluated using single-pass predictive entropy and multi-view TTA mutual information. Exploring learned neural routing heads or feature-space gating mechanisms may further help close the direction predictability gap by learning to discriminate positive corrections from negative degradations directly from feature activations.

6. Conclusion

In this work, we presented a systematic voxel-level characterization of 3D decoder computation in medical image segmentation, evaluating matched full and efficient decoder pairs across diverse backbones, clinical datasets, controlled factor attributions, and two distinct decoder-efficiency mechanisms (capacity removal and a capacity-preserving factorization). Our findings establish a fundamental distinction between two quantities that aggregate metrics conflate: *correctness activity* (where decoder capacity alters predictions) and *directional net gain* (the signed balance of corrections and degradations). Across architectural families and modalities, high-capacity decoders exhibit extensive gross activity that is strongly counterbalanced by bidirectional cancellation, rendering aggregate volume-wide net gain statistically near-zero. Useful decoder computation is highly concentrated ($K_{80} \leq 5\%$) and localized almost exclusively within a narrow 0–2 mm anatomical boundary zone. Crucially, while predictive uncertainty reliably locates *where* computation alters predictions (Location AUROC ≥ 0.865), it fails to discriminate *whether* those alterations improve accuracy (Direction AUROC ≈ 0.520 – 0.591), uncovering a fundamental Activity–Direction Predictability Gap.

These empirical insights motivate a paradigm shift in 3D network design. Traditional dense, symmetric U-Net decoders allocate high-resolution computation uniformly across the volumetric grid, spending substantial compute in organ interiors where its net utility is indistinguishable from zero. Our results argue instead for decoupled, two-stage architectures: a globally lightweight base decoder that generates coarse volume segmentation, coupled to a sparse, conditionally executed high-resolution refinement pathway invoked selectively at ambiguous boundary patches. Realizing this vision requires moving beyond uncertainty-guided spatial proposal to develop true *directional utility estimators* that anticipate expected net accuracy gain.

Clinically, our characterization provides a principled stratification for model deployment rather than a uniform verdict.

Global decoder simplification can be safely adopted for high-throughput screening, intraoperative guidance, and bulk organ volumetry, maximizing inference speed without compromising clinical accuracy. Conversely, heavy decoding capacity should be sparsely targeted to critical anatomical margins—such as tumor margins, vascular walls, and small glandular structures—where small boundary shifts govern high-stakes surgical and radiotherapeutic decisions. By aligning computation with anatomical utility rather than uniform spatial grids, this work provides a foundation for 3D segmentation architectures that are both computationally efficient and clinically dependable.

CRedit authorship contribution statement

Author One: Conceptualization, Methodology, Writing – original draft. **Author Two:** Data curation, Software. **Author Three:** Supervision, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The data used in this study (BTCV/Synapse, MSD01 Brain-Tumour, and MSD08 HepaticVessel) are publicly available.

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Supplementary Material: Qualitative Visualizations for 3D UX-Net and MedNeXt-M-K3

This section provides supplementary qualitative visualizations for 3D UX-Net and MedNeXt-M-K3 backbones, complementing the SwinUNETR qualitative motivation example in Figure 2 of the main manuscript.

Supplementary Material: Entropy Monotonicity with Block-Level Error Rate

Figure S3 visualizes the block-level spatial correlation between mean predictive entropy and block-level error rate across all three evaluated backbones. The monotonic relationship confirms that entropy reliably localizes active, error-prone regions (Location AUROC), even though its directional discrimination ability remains weak.

Supplementary Material: Additional Implementation Details

TTA-MI (Test-Time Augmentation Mutual Information) Setup

For evaluating the predictive power of multi-view epistemic disagreement against spatial entropy, we utilized 8-view Test-Time Augmentation (TTA). Each volume was forwarded through the network across all 8 spatial flip combinations (x, y, z axes). The Mutual Information (TTA-MI) was computed as the difference between the entropy of the mean prediction and the mean of the individual view entropies. Despite this computationally expensive multi-pass estimation, TTA-MI provided only marginally better directional discrimination (Direction AUROC 0.570 for SwinUNETR, 0.539 for 3D UX-Net) compared to single-pass predictive entropy, further reinforcing our conclusion that uncertainty measures are powerful activity indicators but weak directional utility classifiers.

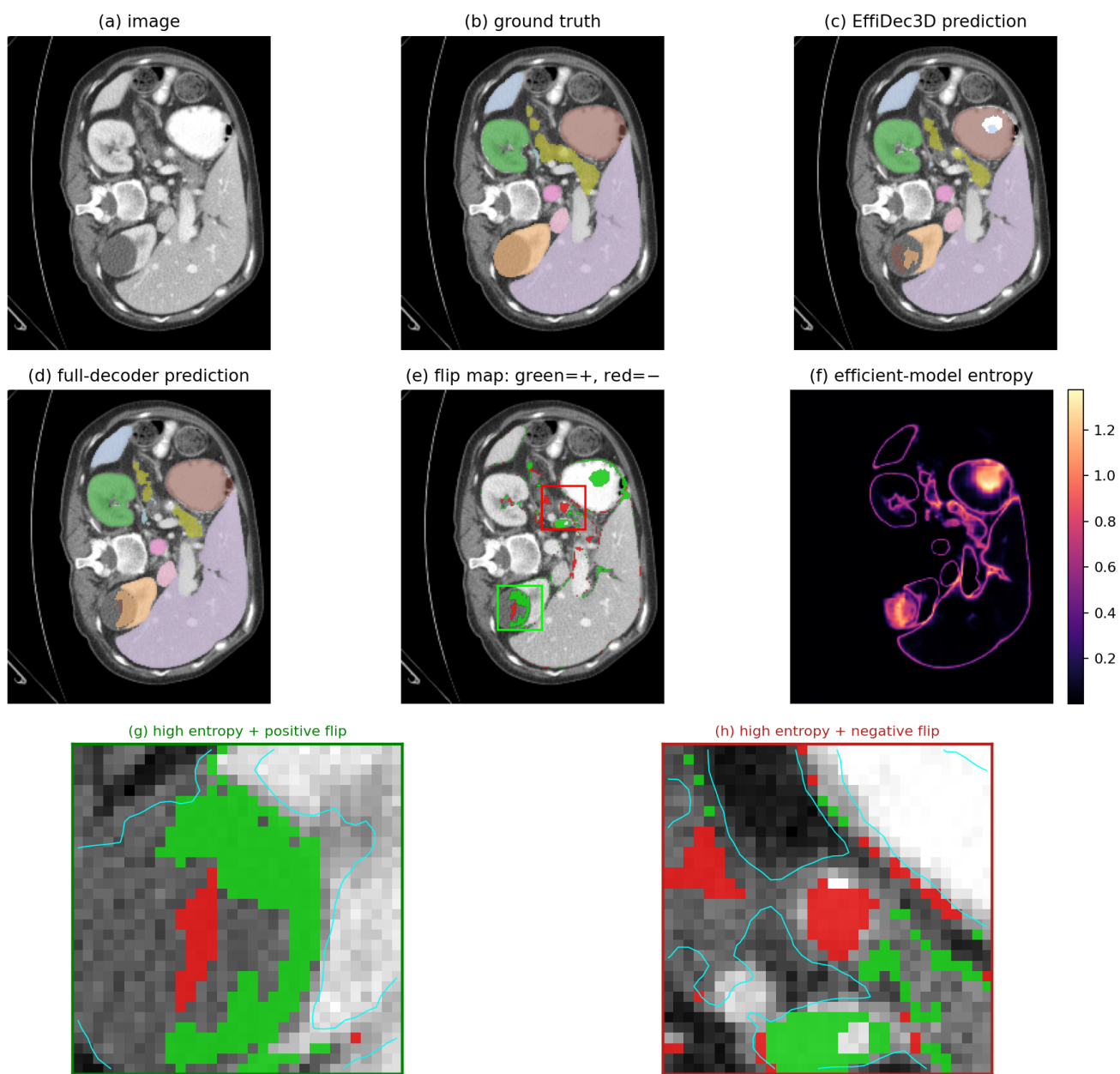


Figure S1: Qualitative motivation visualization for 3D UX-Net on BTCV CT multi-organ dataset under Full (E_0) vs. Efficient (E_1) decoders.

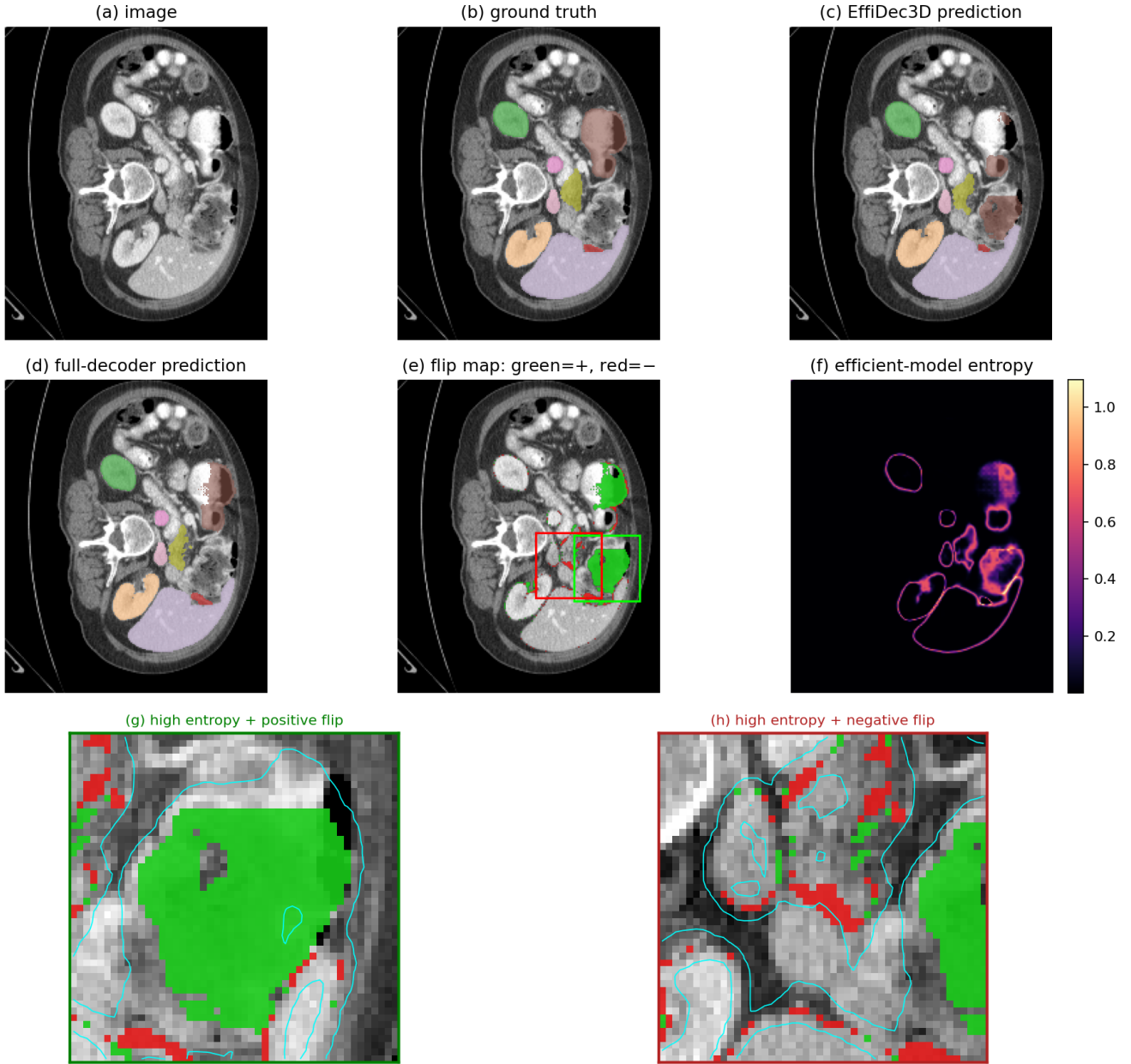


Figure S2: Qualitative motivation visualization for MedNeXt-M-K3 on BTCV CT multi-organ dataset under Full (E_0) vs. Efficient (E_1) decoders.

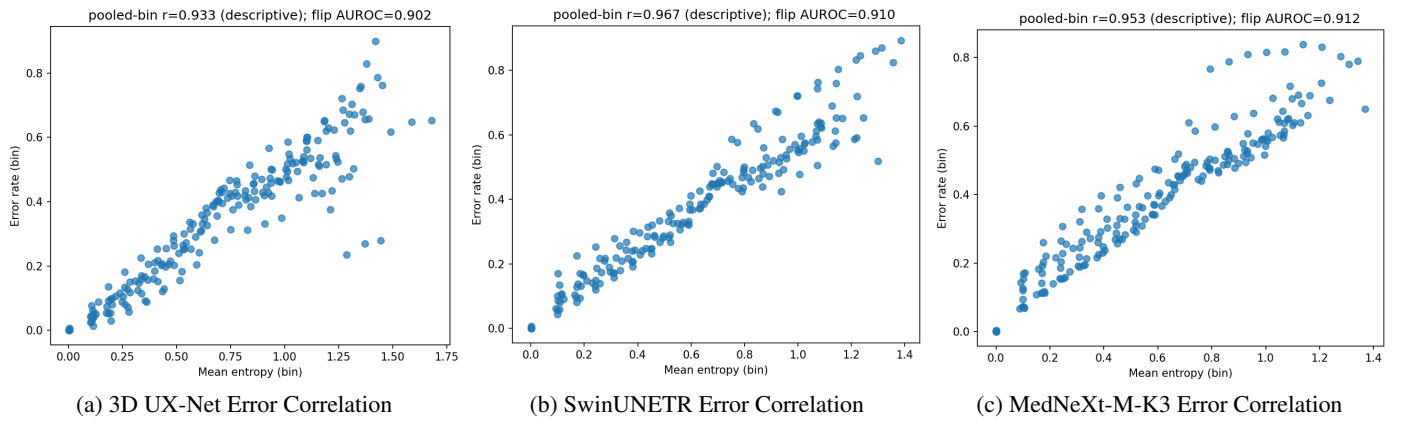


Figure S3: Correlation between mean predictive entropy and block-level error rate across backbones.